Data-Driven Computational Drug Repositioning for *S. aureus*-Associated Diabetic Foot Ulcer

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Introduction

Diabetic foot Ulcer (DFU) is a complication of diabetes mellitus characterized by wounds or ulcers in the lower limbs. The most common pathogen associated with DFU is *Staphylococcus aureus* (SA). Antibiotic resistance and the persistent nature of the pathogen introduce challenges to conventional treatment. Drug repositioning may help determine drugs that could be repositioned for SA-associated DFU (SA-DFU). A diagrammatic illustration of SA-DFU is shown in Figure 1. The following two web resources could be employed for drug repositioning for SA-DFU:

(a) DISGENET (gene-disease associations)

(b) ENRICHR (for determining the related pathways and drugs linked to the disease-associated genes).

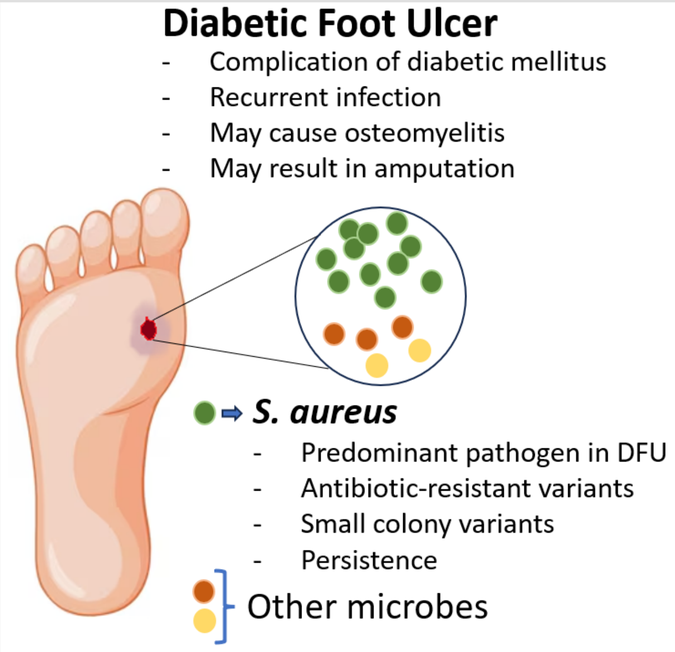

Figure 1. Diagrammatic illustration of SA-DFU

Objective

To identify potential therapeutic drugs for SA-DFU using a computational drug repositioning approach involving gene-disease association analysis and biological pathway enrichment.

Methods

- We utilized DISGENET, a comprehensive database of disease-gene associations to explore genes associated with SA-DFU.
- ENRICHR, a bioinformatics resource was utilized to: (a) perform functional enrichment analysis to identify the KEGG 2021 human biological pathways. (b) determine the drugs that can be enriched from the IDG Drug 2022 database utilizing DISGENET data. A flow diagram of the computational approach is shown in Figure 2.

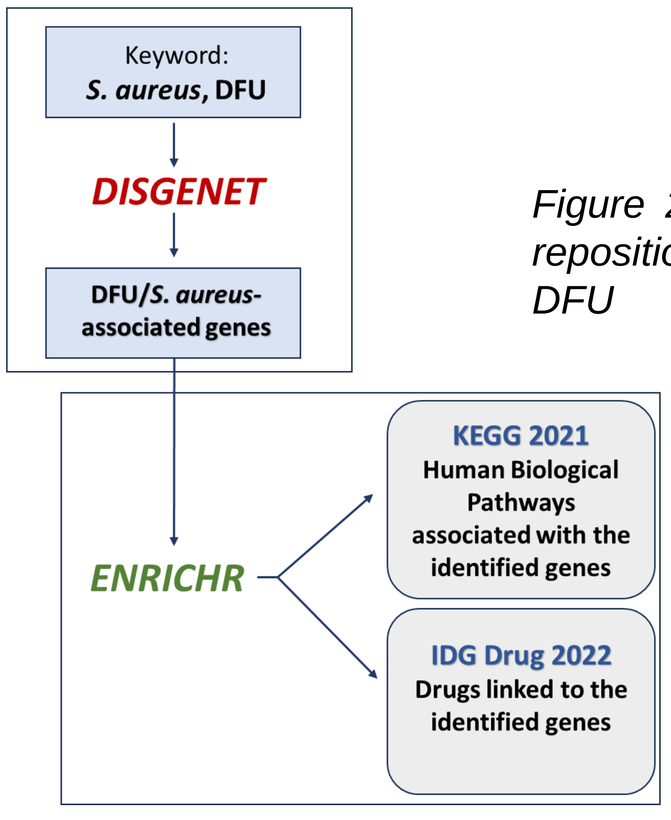

Figure 2: Computational drug-repositioning flow chart for SA-DFU

Results

- We identified 398 genes utilizing DISGENET, which could provide insights into the existing approved drugs that could be repositioned for SA-DFU.
- Table 1 shows the top four KEGG 2021 human biological pathways determined through ENRICHR.
- 132 human biological pathways were identified from KEGG 2021 through ENRICHR with an adjusted P-value (P_{adj}) of <0.05.

Table 1. The top four KEGG 2021 human biological pathways identified through enrichment analysis utilizing ENRICHR

Pathways	Adjusted p-value	Genes
IL-17 signaling pathway	1.3312E-22	CXCL6;CSF3;CSF2;CXCL8;TNF;CXCL5;CCL7;CCL2;MAPK1;MMP1;CCL20;TRAF2;MAPK14;MMP9;MUC5AC;NFKB1;IL4;IL6;IFNG;TRAF3;IL1B;TRAF6;DEFB4A;DEFB4B;S100A8;IL17B;IL17A
Cytokine-cytokine receptor interaction	4.90257E-22	IL22;CXCL6;IL1RN;CSF3;IL20;CSF2;CXCL8;CCL4L1;CSH2;CCL4L2;CSH1;IL24;CXCL14;TNF;CXCL5;TNFSF13B;CXCR1;CCL7;CCL4;CCL2;CCL1;CCR2;IL10;TGFB1;CCL20;GDF15;IFNB1;PPBP;NGF;TGFBR1;TNFRSF1A;IL4;IL1A;IL6;CXCL12;IFNG;IL1B;LEP;CCL28;IL17B;IL17A
Tuberculosis	2.2542E-19	ITGAM;NOD2;TNF;HSPD1;FCGR3A;MAPK1;BID;CAMP;HSPA9;IL10;HLADRB5;TGFB1;VDR;IFNB1;STAT1;MAPK14;NFKB1;TIRAP;TNFRSF1A;TLR1;IL1A;IL6;IFNG;IL1B;TRAF6;TLR9;TLR6;FCGR2B;TLR4;HLA-DRB1;TLR2
Rheumatoid arthritis	2.5195E-19	CXCL6;HLADRB5;CSF2;CXCL8;TGFB1;MMP1;PTH;CCL20;CD80;TNF;CXCL5;TNFSF13B;VEGFA;IL1A;IL6;CXCL12;IFNG;IL1B;CCL2;TEK;TLR4;HLADRB1;TLR2;IL17A

- ENRICHR identified 48 potential drugs with P_{adj}= <0.05 through the IDG Drug 2022 database enrichment for SA-DFU treatment and potentially modulating identified pathways.
- The three significant drugs identified from the IDG Drug 2022 database (Table 2 & Figure 3) were Quercetin (antioxidant and anti-inflammatory), Sorafenib (kinase inhibitor and anticancer agent), and Disulfiram (alcoholism treatment).

Table 2. The top three significant drugs enriched from the IDG Drug 2022 database utilizing ENRICHR

Drug Name	Adjusted p-value	Genes
Quercetin	0.003611718	ACHE;PTPN1;CXCR1;FLT3;PLA2G1B;EPHX2;MMP2;MMP9;ELANE;AURKB;ABCG2
Sorafenib	0.003611718	PLK4;CDK7;FLT3;EPHX2;FLT4;MAPK1;TEK;MAPK14;EPHB1;EPHA3;AURKB
Disulfiram	0.005434397	LOX;CYP1A2;TSPO;OPRM1;SLC6A2;CCR2

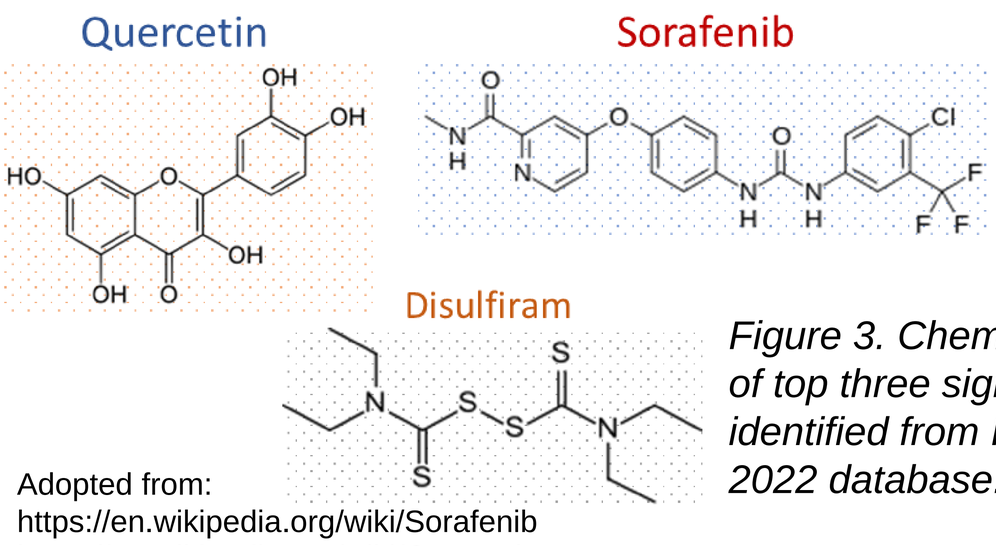

Figure 3. Chemical structure of top three significant drugs identified from IDG Drug 2022 database.

Adopted from:
https://en.wikipedia.org/wiki/Sorafenib
https://en.wikipedia.org/wiki/Quercetin
https://en.wikipedia.org/wiki/Disulfiram

Conclusion

- Further, *in-vitro*, *in-vivo*, and clinical trial studies are necessary to validate the therapeutic potential of identified drugs.
- Computational drug repositioning strategies that combine knowledge from comprehensive gene-disease association studies with biological enrichment analyses to unravel connections between genes to disease pathways and drugs, can be a valuable approach to hasten the discovery of novel therapeutic options for SA-DFUs.